6	**(a)**	packet/quantum/discrete amount of energy	M1	
		of electromagnetic energy/radiation/waves	A1	[2]
	(b)	**(i)** arrow below axis and pointing to right	B1	[1]

(ii)	**1.** $E = hc/\lambda$		
	$= (6.63 \times 10^{-34} \times 3.0 \times 10^{8})/(6.80 \times 10^{-12})$	C1	
	$= 2.93 \times 10^{-14}$ J (*accept 2 s.f.*)	A1	[2]
	2. energy of electron $= (3.06 - 2.93) \times 10^{-14}$		
	$= 1.3 \times 10^{-15}$ J	C1	
	speed $= \sqrt{(2E/m)}$	C1	
	$= 5.4 \times 10^{7}\ \text{m s}^{-1}$	A1	[3]
(c)	momentum is a vector quantity	B1	
	either must consider momentum in two directions		
	or direction changes so cannot just consider magnitude	B1	[2]